# Chapter 3 Solutions

## CH03-Q01

`0x7f + 0x01` produces `0x80`, signed −128. Its mathematical value 128 exceeds 127, so signed overflow occurs.

`0x80 + 0xff` is mathematically −128−1=−129. Its low bits are `0x7f`, signed 127. Signed overflow occurs.

For `0x80 - 0x01`, compute `0x80 + 0xfe + 1`. The low result is again `0x7f`, with signed overflow. Using `0xff` as the complement of `0x01` would be an incorrect subtraction trace.

Signed overflow is $c_{\text{into MSB}} \oplus c_{\text{out of MSB}}$. Unsigned overflow uses only the outgoing carry.

## CH03-Q02

| Round | Old Q0 | Add result C:A | A after shift | Q after shift |
|---|---|---|---|---|
| 1 | 1 | `0:1101` | `0110` | `1101` |
| 2 | 1 | `1:0011` | `1001` | `1110` |
| 3 | 0 | `0:1001` | `0100` | `1111` |
| 4 | 1 | `1:0001` | `1000` | `1111` |

Round 2 shifts full input `1:0011:1101` to low eight bits `10011110`. The carry enters A; it is not accumulated a second time. The final concatenation is `(A,Q)=10001111`, representing $143 = 13 \times 11$.

Fix the initial multiplier $q < 2^n$, and define $R_k = 2^n A_k + Q_k$ after round $k$. The exact invariant is

$$R_k = \left\lfloor \frac{q}{2^k} \right\rfloor + M(q \bmod 2^k)2^{n-k}, \qquad 0 \le k \le n.$$

At $k = 0$, $A_0 = 0, Q_0 = q$, so it holds. When $k < n$, the second term is even. The current low bit is therefore original multiplier bit $q_k$. The conditional addition produces full combined value $R_k + q_k M 2^n$, including the carry. Shifting gives

$$R_{k+1} = \left\lfloor \frac{R_k + q_k M 2^n}{2} \right\rfloor = \left\lfloor \frac{q}{2^{k+1}} \right\rfloor + M(q \bmod 2^{k+1})2^{n-k-1}.$$

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# Chapter 3 Solutions (continued)

The equality uses $q \bmod 2^{k+1} = (q \bmod 2^k) + q_k 2^k$, completing induction. At $k = n$, $q < 2^n$ makes the first term zero, so $R_n = Mq$. A contains the high half and Q the low half. Carry C enters A during the shift and is not lost. The processed old Q0 leaves; old Q1 becomes the next decision bit.

For this case R0..R4 are 11,109,158,79,143, matching the table.

$$T_{\text{serial}} = 3 + 32(1 + 1) + 2 = 69 \text{ cycles.}$$

An initiation interval of four cycles gives $1/4 = 0.25$ multiplication/cycle while a result may still take 69 cycles. Overlap requires separate partial-product/control state, collision handling, and contexts or stage resources for concurrent operations; merely duplicating the ALU is insufficient.

## CH03-Q03

| Round | Shifted A,Q | Trial A−M | State after decision |
|---|---|---|---|
| 1 | `00001,1010` | −4 | Restore, `00001,1010` |
| 2 | `00011,0100` | −2 | Restore, `00011,0100` |
| 3 | `00110,1000` | 1 | Keep and set bit, `00001,1001` |
| 4 | `00011,0010` | −2 | Restore, `00011,0010` |

Thus $Q = 2, R = 3$. Check $13 = 2 \times 5 + 3$ and $0 \le 3 < 5$.

The signed boundary is $-2147483648 / -1 = 2147483648$, above the signed maximum. In traditional MIPS32, `div` does not automatically provide an overflow trap with a dependable quotient. Software must check the operands. Other architectures may define a trap or another result.

## CH03-Q04

$12.5 = 1100.1_2 = 1.1001_2 \times 2^3$. The sign is 1, stored exponent is $3 + 127 = 130 = 10000010_2$, and fraction is `10010000000000000000000`.

```
1 10000010 10010000000000000000000 = 0xc1480000
```

For `0x3f000000`, sign is 0, exponent is 126, and fraction is zero. Value is $1.0 \times 2^{-1} = 0.5$.

The smallest normal has exponent field `00000001` and zero fraction, hence $2^{-126}$. The smallest subnormal has zero exponent field and only fraction bit 0 set, hence $2^{-126}2^{-23} = 2^{-149}$.

# Chapter 3 Solutions (continued)

`0x7f800000` is positive infinity. `0x7fc00001` has all-ones exponent and a nonzero fraction with the quiet bit set, so it is a quiet NaN in this binary32 convention.

## CH03-Q05

Binary32 spacing above 1 is $2^{-23}$. The candidates are $1$ and $1 + 2^{-23}$. The exact sum lies halfway. Candidate 1 has an even low retained significand bit, so nearest-even returns 1, encoding `0x3f800000`.

The addition is inexact. Its flag records that rounding occurred, not the numerical error $2^{-24}$. Trap enablement is a separate control.

A wider intermediate can retain $1 + 2^{-24}$. Subtracting 1 before the final conversion then gives $2^{-24}$, exactly representable in binary32. Rounding the initial addition to binary32 first instead gives zero after subtraction.

## CH03-Q06

$1.5 = 1.1_2 \times 2^0$, while $2^{-126} = 1.0_2 \times 2^{-126}$. Their significand product is $1.1_2$, and their unbiased exponent sum is −126. The product is already normalized, so no left shift is needed.

Stored exponents contain biases. Adding them includes two biases; the result needs one. General multiplication may also require a one-bit right normalization and exponent increment, followed by another adjustment if rounding carries. Special and subnormal cases require separate rules.

## CH03-Q07

Spacing above $2^{24}$ is 2. Adding 1 reaches a midpoint between $2^{24}$ and $2^{24} + 2$. Nearest-even chooses $2^{24}$. Therefore $(x + y) - x = 0$.

The other grouping computes $x - x = 0$, then adds 1, giving 1. The first rounding has lost the unit before cancellation can recover it.

Under the specified wider-intermediate assumption, $(1 + 2^{-24}) - 1 = 2^{-24}$. With binary32 rounding after each operation it is zero. This conclusion depends on retained precision and control settings, not merely on an x87 label.

# Chapter 3 Solutions (continued)

## CH03-Q08

Binary latency is $3 + 32 + 2 = 37$ cycles. Radix-4 latency is $6 + 16(1.5) = 30$ cycle-equivalents. The fractional per-iteration model is an average or abstract timing model, not an assertion that a synchronous stage ends halfway through an ordinary clock cycle.

By the separate acceptance-interval assumption, both throughputs are $1/8 = 0.125$ operations/cycle, or 125 operations per 1000 cycles after filling. No clock frequency is supplied, so operations per second cannot be calculated.

Latency matters for dependent chains and critical paths. Throughput matters when enough independent operations keep resources occupied. A single ordinary iterative unit cannot sustain an eight-cycle interval with a 37-cycle operation without additional overlap or replication.

For a divider, digit selection, normalization, correction, sign handling, zero-divisor checking, and remainder formation remain. Per-iteration timing can differ. Early termination can make costs input-dependent, in which case average time requires an operand distribution.

## CH03-Q09

At 1.0, the retained significand is `1.00000000000000000000000`. Adding $2^{-24}$ leaves GRS=`100`. This is a tie between 1 and $1 + 2^{-23}$; retained LSB zero selects 1, bits `0x3f800000`, with inexact set.

$3 \cdot 2^{-25} = 2^{-24} + 2^{-25}$ gives GRS=`110`. It exceeds half a retained unit, so rounding increments the fraction LSB and yields $1 + 2^{-23}$.

$2^{-151}$ is below half the least positive subnormal $2^{-149}$, so it rounds to +0. $2^{-150}$ is exactly halfway between +0 and $2^{-149}$, selecting even +0, bits `0x00000000`. Both are tiny and inexact and set underflow and inexact.

Maximum finite is `0x7f7fffff`. Adding it to itself exceeds the range; default RNE returns `0x7f800000`, +infinity, with overflow and inexact. `0*infinity` is invalid and defaults to quiet NaN with invalid set, rather than zero.

## CH03-Q10

The encodings are `a=b=0x3f800001`, with exponent 127 and fraction LSB one, and `c=0xbf800002`.

$$ab = (1 + 2^{-23})^2 = 1 + 2^{-22} + 2^{-46}.$$

# Chapter 3 Solutions (continued)

At exponent zero the binary32 spacing is $2^{-23}$. The residual $2^{-46}$ is far below half a ULP, so the separate multiply rounds to $1 + 2^{-22}$. Adding its exact negative gives +0, bits `0x00000000`. The addition cannot recover bits discarded by the multiply.

The fused exact intermediate is

$$(1 + 2^{-22} + 2^{-46}) - (1 + 2^{-22}) = 2^{-46}.$$

This is a normal binary32 value with biased exponent $127 - 46 = 81 = 0x51$, zero sign and fraction, hence `0x28800000`. A single rounding preserves a value lost by the separate product rounding.

At $x = 2^{24}$, ULP=2. `x+1` is a midpoint and rounds to even x, so `(x+1)-x=0`. Computing `x-x` first gives exact zero, then adding 1 gives 1. The different first rounding point causes the discrepancy. Each binary32 operation retains 24 significand bits; real associativity does not bypass these rounding operations.

## CH03-Q11

1. Align a=$1.000_2 \times 2^0$ and b=$1.001_2 \times 2^{-4} = 0.0001001_2 \times 2^0$. The sum is $1.0001001_2$. Retain 1.000 with L=0,G=1,R=0; tail 01 gives S=1. Candidates 1 and 9/8 have midpoint 17/16=136/128. The exact sum 137/128 lies above it, selecting 9/8 with absolute error 7/128. Ignoring S creates a false GRS=100 tie and returns 1, with error 9/128 exceeding half-ULP=1/16.
2. Subtraction gives $0.001_2 \times 2^{-1}$. The left shifts give $0.010_2 \times 2^{-2}$, then $0.100_2 \times 2^{-3}$, then $1.000_2 \times 2^{-4}$. Each equals 1/16. No nonzero bits are discarded, so inexact=0.
3. The candidates for $1.1111_2$ are $1.111_2$ (significand integer 15, odd) and $10.000_2$ (integer 16, even). With L=1 and GRS=100, choose the latter, renormalizing to $1.000_2 \times 2^1$. For $-1.0011_2$, magnitude candidates are $1.001_2$=9/8 (integer 9, odd) and $1.010_2$=5/4 (integer 10, even). Exact magnitude 19/16 is halfway; choose −5/4. Incrementing the magnitude makes this negative result more negative, rather than rounding toward +∞. Both cases are inexact.

# Chapter 3 Solutions (continued)

## CH03-Q12

(1) The exact quotient $2^{-126}/2=2^{-127}=(2^{22})\delta$ has fraction integer $F=2^{22}$ and exponent field 0, giving 0x00400000. It is exact: underflow=0,inexact=0.
For δ/2 the candidates are 0 (F=0, even) and δ (F=1, odd). The tie selects +0, bits 0x00000000. The result is tiny and inexact, setting underflow=1,inexact=1.
For 3δ/2 the candidates are δ (F=1, odd) and 2δ (F=2, even). Select 2δ, bits 0x00000002. The result differs from the exact value, so again underflow=1,inexact=1.
(2) Smallest normal is $2^{23}\delta$ and largest subnormal is $(2^{23}-1)\delta$. Their difference is exactly δ, bits 0x00000001. Both flags remain 0. Small magnitude alone does not determine the underflow flag.
(3) With exponent field 0, the value is $(F/2^{23}) \cdot 2^{-126}=F\delta$. Each unit increase in F adds δ. The last subnormal and first normal also differ by δ, with no enlarged gap. Fixed absolute spacing does not mean fixed relative precision: the same δ is a larger fraction of values nearer zero.

## CH03-Q13

Sign-extend $M = -8$ as `11000` and start with `00000 1101 0`:

| Round | Old $Q_0q_{-1}$ | A action | After add/sub: A Q $q_{-1}$ | After shift: A Q $q_{-1}$ |
|---|---|---|---|---|
| 1 | `10` | −M | `01000 1101 0` | `00100 0110 1` |
| 2 | `01` | +M | `11100 0110 1` | `11110 0011 0` |
| 3 | `10` | −M | `00110 0011 0` | `00011 0001 1` |
| 4 | `11` | 0 | `00011 0001 1` | `00001 1000 1` |

Discard the history bit: $(A, Q)$ =`00001 1000`. Remove the redundant sign bit to get `00011000`, or 24. The multiplier recodes as $-1 + 2 - 4 = -3$, giving $8 - 16 + 32 = 24$.

With only four bits, the first round requires $0 - (-8) = 8$, but `1000` means $-8$. The incorrect trace is:

| Round | Old $Q_0q_{-1}$ | A action | After add/sub: A Q $q_{-1}$ | After shift: A Q $q_{-1}$ |
|---|---|---|---|---|
| 1 | `10` | −M | `1000 1101 0` | `1100 0110 1` |
| 2 | `01` | +M | `0100 0110 1` | `0010 0011 0` |
| 3 | `10` | −M | `1010 0011 0` | `1101 0001 1` |
| 4 | `11` | 0 | `1101 0001 1` | `1110 1000 1` |

# Chapter 3 Solutions (continued)

Final `11101000` means $-24$, not 24. Signed overflow occurs in the first round; later arithmetic shifts cannot restore the sign information already lost.

For `0101`, low-to-high digits are $(-1,+1,-1,+1)$: $-1+2-4+8=5$, requiring four arithmetic operations. Unsigned shift-add sees only two ones and needs two additions. Four rounds do not establish four cycles without timing rules for arithmetic, shifting and control.

Six-bit $-13$ is `110011`. Append history0. The overlapping groups are $(q_1q_0q_{-1})$=`110`, $(q_3q_2q_1)$=`001`, $(q_5q_4q_3)$=`110`. Digits $(-1,+1,-1)$ give $-13=-1+4-16$.

| j | Digit | Partial product | Twelve-bit representation |
|---|---|---|---|
| 0 | −1 | +5 | `000000000101` |
| 1 | +1 | −20 | `111111101100` |
| 2 | −1 | +80 | `000001010000` |

The sum is $5-20+80=65$, or `000001000001`. Substitute $d_{2j}=q_{2j-1}-q_{2j}$ and $d_{2j+1}=q_{2j}-q_{2j+1}$ into $c_j=d_{2j}+2d_{2j+1}$. This yields $c_j=q_{2j-1}+q_{2j}-2q_{2j+1}$, which determines all eight table entries.

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## CH03-Q14

### 1. Check the total at each level

| Row | Eight bits | Unsigned value |
|---|---|---|
| $P_0$ | 00001011 | 11 |
| $P_1$ | 00010110 | 22 |
| $P_2$ | 00101100 | 44 |
| $P_3$ | 01011000 | 88 |
| $S_1$=$P_0$ XOR $P_1$ XOR $P_2$ | 00110001 | 49 |
| $C_1$=majority($P_0$,$P_1$,$P_2$) shifted left | 00011100 | 28 |
| $S_2$=$S_1$ XOR $C_1$ XOR $P_3$ | 01110101 | 117 |
| $C_2$=majority($S_1$,$C_1$,$P_3$) shifted left | 00110000 | 48 |

Both 49+28+88 and 117+48 equal 165=11×15. Majority is bitwise; it does not consume a neighboring carry.

# Chapter 3 Solutions (continued)

## 2. Final carries and the missing-shift error

| Position j, low to high | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $S_2[j]$ | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 |
| $C_2[j]$ | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| Carry-in $c_j$ | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| Result bit | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| Carry-out $c_{j+1}$ | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |

Read high to low: $10100101_2=165$. Omitting both left shifts gives erroneous first-level rows 49 and 14, then erroneous second-level rows 103 and 24 from inputs 49,14,88. Their sum is 127. The bug halves the carry's weight.

## 3. Sum the one-column identities

For a bit sum 0,1,2 or 3, $x_j+y_j+z_j=s_j+2k_j$. Multiply by $2^j$ and sum over j=0,...,W−1. This proves X+Y+Z=S+C, including position W of C. Three four-bit values 15 give $S=1111_2=15$,$C=11110_2=30$ and exact total $101101_2=45$, requiring six bits. In four-bit modular arithmetic, truncate C to $1110_2=14$ and obtain (15+14) mod 16=13=$1101_2$. The exact maximum is $3(2^W-1)$, which can require W+2 bits for $W\geq 2$. For W=1 the maximum is 3 and needs only two bits.

## 4. Count levels before evaluating delay

Nine rows become 6,4,3,2: four levels. The stated upper-bound model gives $40+4\times 60+180+20=480$ ps. This is a model evaluation, not a measurement or a comparison against an unspecified competing circuit. Such a comparison needs its adder, wiring, load and technology data.

# Chapter 3 Solutions (continued)

## 5. All five independent transactions

| Transaction | Accept edge | Accept time/ps | Complete edge | Complete time/ps |
|---|---|---|---|---|
| A | 0 | 0 | 3 | 750 |
| B | 1 | 250 | 4 | 1000 |
| C | 2 | 500 | 5 | 1250 |
| D | 3 | 750 | 6 | 1500 |
| E | 4 | 1000 | 7 | 1750 |

Latency is 750 ps, initiation interval 250 ps and final completion 1750 ps. If B consumes A's product, it lacks an operand at edge 1 and must be delayed according to the actual interface and sampling constraints. The independent-transaction assumption fails; the CSA arithmetic remains valid.

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# Chapter 3 Solutions (continued)

## Verification code appendix: ieee754_explorer.py

```
import math
import struct


def bits_to_f32(bits: int) -> float:
    return struct.unpack(">f", struct.pack(">I", bits))[0]


def f32_to_bits(value: float) -> int:
    try:
        return struct.unpack(">I", struct.pack(">f", value))[0]
    except OverflowError:
        return 0xFF800000 if value < 0 else 0x7F800000


def decode_f32(bits: int) -> dict[str, int | float | str]:
    sign = bits >> 31
    exponent = (bits >> 23) & 0xFF
    fraction = bits & 0x7FFFFF
    if exponent == 0xFF:
        kind = "infinity" if fraction == 0 else "NaN"
    elif exponent == 0:
        kind = "zero" if fraction == 0 else "subnormal"
    else:
        kind = "normal"
    return {
        "sign": sign,
        "exponent_field": exponent,
        "fraction_field": fraction,
        "kind": kind,
        "value": bits_to_f32(bits),
    }


def round_nearest_even(guard: int, round_bit: int, sticky: int, lsb: int) -> bool:
    """Return whether the retained significand increments."""
    return bool(guard and (round_bit or sticky or lsb))


def main() -> None:
    assert f32_to_bits(-12.5) == 0xC1480000
    assert decode_f32(0x3F000000)["value"] == 0.5
    assert decode_f32(0x00000001)["kind"] == "subnormal"
```

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# Chapter 3 Solutions (continued)

```
    assert decode_f32(0x00000001)["value"] == math.ldexp(1.0, -149)
    assert decode_f32(0x00800000)["value"] == math.ldexp(1.0, -126)
    assert decode_f32(0x7F800000)["kind"] == "infinity"
    assert decode_f32(0x7FC00001)["kind"] == "NaN"

    rounded_once = bits_to_f32(f32_to_bits(1.0 + math.ldexp(1.0, -24)))
    assert rounded_once == 1.0

    x = float(2**24)
    left_grouping = (bits_to_f32(f32_to_bits(x + 1.0)) - x)
    right_grouping = (x - x) + 1.0
    assert left_grouping == 0.0
    assert right_grouping == 1.0
    assert left_grouping != right_grouping

    assert round_nearest_even(guard=1, round_bit=0, sticky=0, lsb=0) is False
    assert round_nearest_even(guard=1, round_bit=1, sticky=0, lsb=0) is True

    a = bits_to_f32(0x3F800001)  # 1 + 2^-23
    c = bits_to_f32(0xBF800002)  # -(1 + 2^-22)
    separate_product = bits_to_f32(f32_to_bits(a * a))
    assert f32_to_bits(separate_product) == 0x3F800002
    assert separate_product + c == 0.0
    assert f32_to_bits(math.ldexp(1.0, -46)) == 0x28800000

    assert f32_to_bits(math.ldexp(1.0, -150)) == 0x00000000
    assert f32_to_bits(bits_to_f32(0x7F7FFFFF) * 2.0) == 0x7F800000

    print(f"-12.5 bits: 0x{f32_to_bits(-12.5):08x}")
    print(f"smallest subnormal: {decode_f32(0x00000001)['value']:.17e}")
    print(f"1 + 2^-24 rounds to: {rounded_once}")
    print(f"grouping comparison: {left_grouping} != {right_grouping}")
    print(f"separate mul+add: 0x{f32_to_bits(separate_product + c):08x}")
    print(f"fused exact result: 0x{f32_to_bits(math.ldexp(1.0, -46)):08x}")
    print("IEEE-754 checks: PASS")


if __name__ == "__main__":
    main()
```

### References and attribution

David A. Patterson, John L. Hennessy. Computer Organization and Design: The Hardware/Software Interface, Revised Fourth Edition.

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